Mobile video platform for diagnosing human motor anomalies

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Abstract - The growing demand for telemedicine services necessitates innovative solutions for monitoring patients with movement disorders. The proposed mobile video surveillance system integrates computer vision, deep learning, and intelligent analysis methods to diagnose human movement anomalies accurately in real-time. The system's key components are algorithms for detecting and tracking human movements, contextual analysis, and a decision-making system. The system makes decisions about the detection or possibility of incidents, such as falls, and sends notifications to users via a mobile application. The integration of IoT sensors enables the processing of physiological data (heart rate, activity), which increases the accuracy of diagnostics and expands the system's range of applications. The proposed modular architecture of the system provides scalability and adaptability to different operating conditions. The system meets privacy standards through the use of encryption and multi-level authentication.
Our proposed system ensures effective monitoring of patients with motor disorders, detects incidents promptly, and sends alerts, improving the quality of telemedicine services.

KEYWORDS
mobile health, patient monitoring, diagnostics of human movement abnormalities, computer vision, video stream processing, contextual analysis, intelligent analysis

1. Introduction

1.1 Rationale for the development of a mobile video platform for the diagnosis of motor anomalies

The modern healthcare system faces numerous challenges, such as an aging population, an increase in chronic diseases, including those associated with impaired motor functions, and a growing demand for remote monitoring and telemedicine services [1]. Innovative health monitoring technologies based on the analysis of patient's movement patterns, especially using mobile platforms, are an important area of medical development, as they allow monitoring patients from a distance and automatically diagnosing and reporting abnormalities in movement [2].

Scientific approaches to physical condition monitoring using artificial intelligence (AI) and computer vision technologies demonstrate significant potential for automated anomaly detection. For example, the wearable devices and sensors reviewed by Patel et al.[3] are actively used in rehabilitation programs to assess patients' physical parameters. However, such devices are limited in functionality and cannot fully provide the reliable video monitoring required to diagnose complex movement anomalies. Similar limitations are also inherent in other systems: for example, MediaPipe [4], widely used to track posture and movements, does not allow detection of subtle deviations in exercise performance, which are critical for accurate diagnosis and rehabilitation.

Our proposed mobile video platform system for detecting movement anomalies (Figure 1) differs from existing solutions' ability to integrate multiple technologies. It combines segmentation and precise motion monitoring capabilities based on deep learning, computer vision technologies, and mobile platforms, allowing for real-time data acquisition and continuous monitoring of the patient's physical condition. Unlike existing platforms that focus on individual health indicators, this system provides comprehensive monitoring, which allows for the detection of hidden patterns and movement anomalies that are critical for the complete rehabilitation of patients [5].

Thus, the main goal of this work is to propose a new approach to creating an integrated mobile platform for tracking and diagnosing movement anomalies, which includes a modular structure that can connect additional sensors and IoT devices. This solution creates new opportunities for improving the healthcare system due to its high accuracy, adaptability, and expandability.

Fig. 1. Illustration of the use of a video platform for the diagnosis of motor anomalies

1.2 Justification for the choice of technologies

The choice of innovative technologies for developing a mobile video platform to diagnose movement anomalies is based on the analysis of current trends in digital medicine and the capabilities of computer vision and deep learning. Given the growing need for remote monitoring and telemedicine services, such technologies are highly relevant. One of the key components is computer vision algorithms, such as Vision Transformers (ViT)[6], which allow analyzing video streams with high accuracy and can be used to recognize motion patterns and detect anomalies in real-time.

The MediaPipe Pose Estimation technology [7] is also an important tool, as it allows for accurate tracking of key points on the body, such as joints and limbs, which is critical for diagnosing movement anomalies. Using MediaPipe combined with deep learning algorithms provides the system with high sensitivity to the slightest changes in patient movements, which increases its effectiveness in rehabilitation programs. In addition, the Segment Anything Model (SAM)[8] technology allows you to automatically highlight body parts in videos, which is necessary to control the accuracy of exercises.

Integration with IoT devices ensures continuous data collection, making it possible to create comprehensive systems to monitor the user's physical performance in real-time [9]. For example, IoT sensors can record vital signs such as heart rate, activity level, or body position. Thus, the collected data from different sources is combined into a single stream, which allows for the accurate detection of hidden patterns and potential health problems.

The proposed system is characterized by its adaptive architecture, ensuring flexibility and scalability. The modular structure allows adding new components or connecting additional devices without significant changes to the overall architecture of the platform [10]. This approach ensures the system can be constantly updated and expanded in response to new market requirements.

All of these technologies enhance the mobile video platform system, which provides effective control over users' physical condition and improves the accuracy of diagnosing motor anomalies.

2. Methodology

2.1 Architecture of the data collection and processing system

The developed mobile video platform for detecting movement anomalies is based on modern computer vision, deep learning, and big data processing technologies. It involves a multi-level process of data collection, processing, and analysis, which allows the detection of anomalies in patient behavior in real-time, taking into account the context and specific environmental conditions. The main steps of the process of collecting, processing, and outputting information, as well as the key components of the architecture that implement the functionality of the proposed platform, are described below.

Step 1: Integrate indoor cameras and the mobile app

1. Stationary cameras:
   Installing high-resolution cameras in controlled areas (e.g., hospitals or nursing homes) ensures constant monitoring and the ability to continuously transmit video streams to a server for further analysis [11]. Each camera has a stable network connection, providing reliable data transmission to the processing server.
2. Mobile application:
   The mobile app acts as an interface for healthcare professionals and relatives, providing data visualization, real-time alerts, and access to event history. The application is integrated with the server via the REST API, which ensures instant data synchronization and sends push notifications in case of critical events, such as falls or abnormal movements [12].
3. Central data processing server:
   Based on FastAPI or Flask, the server acts as a central node that receives video streams, coordinates data processing, and interacts with a decision-making model (Llama 3.2) to analyze falls. The server gets real-time video frames, processes them, and manages data storage for further analysis [13].

Step 2: Detecting people and estimating body position

1. Video stream processing and frame extraction:
   The system is optimized to allocate video frames at regular intervals (for example, every 0.5 seconds), which reduces the computational load while maintaining high analysis speed. This is achieved by using a frame buffering algorithm on the server.
2. Detecting people with YOLO:
   To identify faces in a video stream, the YOLO model [14] (e.g., YOLOv8) is used to find people in each frame and select bounding boxes for

further processing. YOLO works effectively with dynamic scenes and minimizes false positives in cases with many objects.

3. MediaPipe for assessing body posture and position:

   For each person detected by YOLO, MediaPipe is used to identify key body points such as shoulders, knees, and hips. MediaPipe can identify abnormal postures and rapid movements that could indicate falls or other potentially dangerous events. Using these key points, the system flags as "fall candidate" footage where unusual postures or rapid changes in movement are detected.

Step 3: Contextual analysis with segmentation and action recognition

1. Segment Anything Model (SAM):

   For footage marked as a "fall candidate," the SAM model is used to accurately segment the detected person in the context of the overall scene. This allows for a more detailed outline of the body shape, even when the person is close to other objects, such as furniture or medical equipment, reducing false positives.

2. Contextual analysis with CLIP:

   To improve scene interpretation, the CLIP[15] model is used, which analyzes frames with "fall candidates" to determine the context of the situation. For example, CLIP can help recognize whether a person is near a staircase, bed, or other object that potentially increases the risk of falling. This allows for objects in the environment to be considered and contextual data to be added to the analysis to minimize false positives.

Step 4: Making decisions about falling with Llama 3.2

1. Data aggregation for Llama 3.2:

   The system collects key parameters for analysis using the Llama 3.2 model[16]: bounding box coordinates, key body position points, segmentation masks, and contextual data obtained from CLIP. This data is structured into a query sent to Llama 3.2 to decide the probability of falling. For example, the query may look like this:

"A person was found lying near the stairs after a sudden change in posture. Is this likely to be a fall?" Llama 3.2 processes the data and returns a probabilistic estimate or textual description of whether a fall has occurred [9].

2. Activation of notifications based on Llama analysis 3.2:

   If Llama 3.2 estimates a high fall probability, the server generates a notification containing a brief description of the event, for example, "Fall detected near stairs." This notification and a snapshot of the footage are transmitted to the mobile application.

Step 5: User interface and notification management via the mobile app

1. Display notifications:
The mobile app provides an interface for viewing alerts, which displays the frame, location of the incident, and probability of fall. Users (e.g., caregivers or medical professionals) can view the alert, acknowledge it, or cancel it, allowing feedback to further improve the model's accuracy.
2. Push notifications and integration with telemedicine:
Notifications are sent to guardians or emergency contacts with a brief event description. The data can be automatically transferred to telemedicine systems for immediate intervention or remote consultation if the event is confirmed.

Step 6: Logging events and continuous improvement

1. Event logging and feedback processing:
All events are stored in logs, including frames, Llama 3.2 responses, and user feedback. This allows you to analyze model accuracy and use the collected data to train or retrain models to improve accuracy.
2. Improving the model:
The system can adapt thresholds for detecting falls and other parameters based on user feedback. Additional conditions or refinements to the scene analysis can improve accuracy and adjust the model to new operating conditions [13].

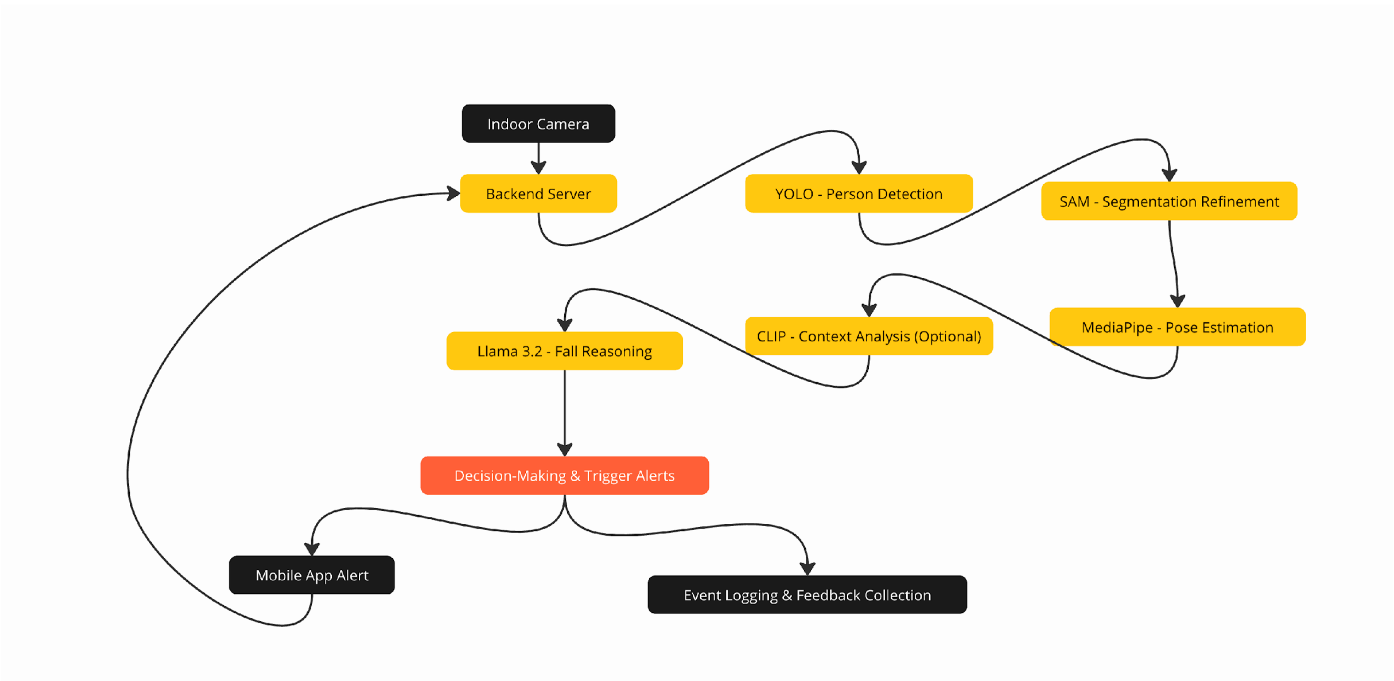

Fig. 2. Pipeline video platform for the diagnosis of motor anomalies

The video platform for the diagnosis of motor abnormalities, shown in Figure 2, includes the following technologies:

- Video processing and object detection: YOLOv8 for face detection, MediaPipe for body pose estimation.
- Segmentation and contextual analysis: SAM for object segmentation, CLIP for contextual classification (optional).
- Decision-making: Llama 3.2 to analyze data and conclude the falling probability.
- Server processing: FastAPI or Flask for frame processing, notification management, and mobile app interaction.
- Mobile application: A platform for receiving notifications, user interaction, and feedback management.

This architecture provides a flexible and accurate process for monitoring user health, combining computer vision, deep learning, and contextual analysis technologies to achieve high reliability and adaptability of the system.

2.2 Conceptual architecture of a mobile video platform for the diagnosis of motor anomalies

The conceptual architecture of the mobile video platform for diagnosing movement anomalies is based on a combination of several integrated modules, each of which performs specific functions in the system's overall structure. This approach ensures a high level of scalability, the ability to incorporate additional modules, and adaptation to different use cases. Below is a detailed description of the leading architectural components and levels of interaction.

The primary levels of conceptual architecture:

- **Level of data collection**
  - **High-resolution cameras** provide continuous video surveillance in controlled areas such as patient rooms or medical offices. The cameras are connected to a network and transmit data in real-time to a processing server [17].
  - **IoT sensors** complement cameras by recording critical physiological parameters of the user, such as heart rate and activity level, which allows the system to perform comprehensive monitoring [18].
- **Level of data pre-processing**

    - **Face detection module (YOLO)**: The YOLO algorithm identifies the presence of faces in the frame and creates bounding boxes around each detected object. This reduces computational overhead because further analysis is focused only on the selected areas of the frame.
    - **Posture assessment with MediaPipe**: After detecting the faces in the frame, MediaPipe identifies the main key points of the body (joints, limbs) and performs a fundamental posture analysis, which allows you to identify anomalies in body position and mark potential falls.
- **Level of contextual analysis**
    - **Segment Anything Model (SAM)** performs segmentation of frames where potential falls are marked. This provides more detailed identification and allows you to distinguish body contours from the background or other objects. SAM improves accuracy in environments with multiple objects, reducing the number of false positives.
    - **Contextual analysis with CLIP**: CLIP performs a contextual analysis of the scene to determine if a person is near potentially dangerous objects (stairs, furniture, etc.). This helps to consider the environment to detect falls more accurately, increasing the solution's reliability.
- **The level of decision-making based on AI**
    - **The Llama 3.2 model** receives structured data from the previous levels: bounding boxes, key points of body position, segmented masks, and contextual information. Based on this, a query is generated for the model, which contains a description of the detected situation and the probability of a fall. Llama 3.2 analyzes this information, returning a likelihood or a textual conclusion about the possibility of an incident.
    - **Automatic notification**: If Llama 3.2 detects a high probability of a fall, the system generates an alert. This alert is sent to the mobile app, allowing users to view the incident footage and relevant details.
- **Level of user interface and feedback system**
    - **The mobile app** serves as the primary interface for viewing alerts in real-time. Users can view incident footage, confirm or cancel alerts, and provide feedback. This allows data to be collected to improve the system further.
    - **Push notifications and integration with emergency services**: The app supports sending push notifications to emergency contacts and medical personnel when a critical incident is confirmed. This makes it possible to respond quickly to falls or other dangerous situations.
- **Level of continuous improvement**
    - **Event logging and data analysis**: All events, including Llama 3.2 responses and user feedback, are stored in logs, allowing you to analyze system accuracy, reduce false positives, and optimize processing algorithms.

    - **Adaptation of model parameters**: Based on user feedback, the system dynamically adjusts parameters such as thresholds for anomaly detection, which allows it to adapt to new operating conditions and improve system accuracy continuously [19].

2.3 Integration with mobile applications and IoT devices

Integrating a mobile video platform with mobile applications and IoT devices is key to creating a comprehensive approach to diagnosing movement anomalies. This integration provides continuous monitoring, data collection from multiple sources, and the ability to respond immediately to critical situations. Below, we describe the integration mechanisms and their role in ensuring the platform's efficiency.

- **Communication with the mobile application**:
    - The mobile application is the primary means of interaction between the system and users, including healthcare professionals or family members of patients. The application is synchronized with the processing server via an API (e.g., REST API), allowing you to receive and transmit data quickly.
    - App users can receive real-time notifications of detected falls or movement anomalies, view footage of incidents, and confirm or cancel alerts. This feedback allows the system to improve its accuracy based on real-world feedback.
- **Integration of IoT sensors**:
    - IoT sensors like heart rate monitors, pedometers, and accelerometers are connected to the system via specialized data transfer protocols such as MQTT or HTTP. These sensors transmit the user's physiological parameters to the processing server, combining them with video data to create a complete picture of the patient's physical condition [20].
    - Physiological indicators obtained from IoT sensors add a layer of contextual analysis, allowing the system to detect motor anomalies and assess the user's overall health. For example, a combination of low motor activity and an elevated heart rate can signal a critical situation, such as a heart attack or other cardiovascular problems [21].
- **Data transmission infrastructure**:
    - The system uses protocols that support high-speed transmission and low latency to ensure continuous data exchange. MQTT ensures reliable data transmission even in case of poor connection, as the protocol can store messages until the connection is restored. HTTP provides stable and secure data exchange, especially for transmitting physiological indicators [22].

    - Thanks to these technologies, the platform can support continuous real-time monitoring, which is especially important for immediate response to potential incidents.
- **Notification and feedback system**:
    - The notification system is designed to inform users about critical events in real-time. Push notifications are sent to the mobile application with a description of the incident and appropriate recommendations. For example, the notification may contain the text "A fall was detected near the stairs" along with a snapshot of the incident, which allows for a quick assessment of the situation.
    - In addition, users can provide feedback directly through the app, which allows them to confirm or deny the detected incident. This feedback is sent back to the processing server, which is used to improve the algorithms further and reduce the number of false positives.

Integrating mobile applications and IoT devices significantly increases the video platform's efficiency, allowing users to monitor and analyze patients' conditions in real-time. This approach also increases the convenience and accessibility of the system for medical staff and patient's relatives, which positively impacts the quality and timeliness of services provided.

2.4 Notification and Feedback System

When a motion anomaly is detected, the notification and feedback system automatically generates an alert sent to the user's mobile application. The instant messaging system ensures immediate notification of responsible persons, critical for timely response to potentially dangerous incidents.

Interactive feedback in the mobile app allows users to confirm or cancel alerts, significantly improving the system's accuracy, especially in reducing the number of false positives. This information is used to train recognition algorithms.

In cases where the seriousness of the incident is confirmed (for example, if the fall resulted in an injury), the system automatically initiates an escalation, sending a notification to the specified emergency contacts.

The alert system also provides analytics and reporting capabilities. Regular analysis of feedback allows the platform to improve the accuracy of critical event detection by adapting thresholds or adjusting parameters for specific monitoring conditions.

2.5 Data protection and privacy

Data protection and user privacy are key aspects of the mobile video platform used to diagnose movement anomalies. To ensure the safety of personal and medical data, modern encryption technologies and multi-level security mechanisms are used to guarantee the secure exchange of information between different platform components.

The platform implements data encryption both during transmission and storage. Data is transferred between devices via secure communication protocols such as HTTPS and MQTT with TLS support, which protects data from interception during transmission. All video and medical data collected by the platform is encrypted using AES-256 algorithms, a standard for protecting confidential information, and complies with international security standards [23].

Access to the system is restricted through multi-level authentication, including two-factor authentication (2FA) for users with administrative privileges and healthcare professionals. This ensures protection against unauthorized access, particularly in device loss or theft. In addition, the system periodically updates access tokens to provide an additional layer of protection against malicious attacks.

Privacy is also maintained through role-based access restriction policies. This segmentation allows you to provide access only to the necessary information, reducing the risk of a privacy breach.

The system also complies with international data protection standards, including the GDPR (General Data Protection Regulation of the European Union) and HIPAA (Health Insurance Portability and Accountability Act of the United States). This ensures that users' privacy rights are respected, as well as the platform's obligation to report security incidents [19].

Thus, the platform utilizes advanced security and privacy practices to ensure the safety of user data at all levels. All measures, including encryption, multi-level authentication, access restriction policies, and compliance with international standards, make the platform reliable and protected from external threats.

2.6 Implementation of the system at the level of subsystem interaction

The implementation of the mobile video platform system for the diagnosis of motor anomalies includes the integration of subsystems within the framework of a pilot project that demonstrates the ability of various components to interact and coordinate to achieve high diagnostic accuracy. The pilot implementation is designed to test the effectiveness of the main modules at the integration level by simulating the interaction between them.

At the video data processing level, integration with pose detection and estimation algorithms such as YOLO and MediaPipe is used. The test video frames are transferred to the server, where these algorithms detect faces and estimate their positions.

For a more detailed analysis of the subsystem's interaction, a segmentation module was implemented using the SAM model, which provides outline detection of objects in the frame. Based on the data obtained, the contextual analysis module, built using the CLIP model, identifies objects around the patient, including potentially dangerous objects such as stairs or a bed. This helps to improve accuracy and reduce the risk of false positives by considering the context of each situation.

As part of the pilot project, we also implemented interaction with the decision-making system based on Llama 3.2, enabling probabilistic event estimates to form. The system aggregates data from the previous modules and generates structured queries to Llama 3.2, which interprets the data and returns an estimate of the probability of a critical event.

The pilot project's alert system integrates with a mobile application where users can receive notifications of potential incidents in real-time.

Thus, the implementation of the pilot project demonstrates the effectiveness of the interaction of subsystems and confirms that the proposed system can support the actual operation of a mobile video platform for the diagnosis of motor anomalies. This creates an opportunity for further scaling the system, including integrating additional functions and modules into a full version of the platform for clinical use.

3. Results

The proposed mobile video platform system for diagnosing motor anomalies demonstrates significant potential through the use of the latest computer vision methods, deep learning, and the integration of automated contextual analysis. The system's primary goal is to provide accurate and continuous monitoring of the patient's physical condition, including analysis of movement patterns, detection of movement anomalies, and timely response to critical events.

The system is built on the step-by-step integration of various technological components. At the first level of video data processing, the system uses the YOLO object detection model to identify the patient and build bounding boxes, significantly increasing subsequent processing steps' efficiency and accuracy. Once an object is detected, MediaPipe's algorithm identifies key points on the body, such as the position of the shoulders, joints, and limbs, allowing the platform to track overall movement and accurately identify potentially dangerous poses or anomalies.

The next level of processing is based on the Segment Anything Model (SAM) for object segmentation. The SAM model can create a detailed mask for each detected object, which allows it to correctly recognize the patient's body shape, even if other objects are in the frame, such as furniture or other people.

The system features contextual analysis based on the CLIP model. Using CLIP makes it possible to recognize objects around the patient and interpret the environment. For example, suppose the patient is near stairs or in a high-risk area (bathroom, kitchen). In that case, the system increases the priority of notifications for such cases, as a fall in these conditions can have serious consequences. This approach provides a more accurate analysis that considers the patient's posed movements and the context of their position in space.

The key component of the decision-making system is the Llama 3.2 model, which analyzes the collected data from various sources, including information about body position, physiological parameters, and contextual data about the environment received from CLIP. Based on this data, the Llama 3.2 model generates probabilistic event estimates, allowing the platform to adapt to new scenarios and specific conditions. The ability of Llama 3.2 to process structured queries and analyze various indicators provides the platform with a high level of reliability and adaptability.

The last essential component is the notification and feedback system, which ensures interactivity and allows users to respond quickly to incidents. Interactive feedback in the mobile application will enable confirmation or cancellation of alerts, which significantly improves the accuracy of the platform, in particular by reducing the number of false positives.

The proposed system has demonstrated high efficiency by combining computer vision, deep learning, and interactive feedback technologies to provide an accurate, adaptive, and scalable solution for diagnosing movement anomalies. Thanks to its versatile modular architecture, the platform is suitable for use in various medical and social settings where accurate and timely patient monitoring is required.

4. Discussion

4.1 Importance for the healthcare system

Our proposed mobile video platform for diagnosing movement abnormalities has significant potential to improve the healthcare system, as it offers an innovative approach to remote patient monitoring that ensures timely and continuous care. Such a platform is handy for healthcare facilities caring for elderly patients with an increased risk of falling or limited mobility, as it provides additional safety and control over their physical condition without the need for constant medical staff presence.

An important aspect is the platform's ability to contribute to developing a patient-centered healthcare system. This is achieved through the ability to integrate the platform into patients' homes, allowing them to live safer at home, especially for patients with limited mobility.

Thus, introducing this platform can be essential in developing remote monitoring and telemedicine, particularly for patients with chronic diseases, the elderly, and people with special needs. Thus, the platform provides a safer, more efficient, and more convenient environment for patient care while reducing the cost of running medical facilities.

4.2 Challenges and limitations

Despite its considerable potential, the proposed mobile video platform for diagnosing movement anomalies faces several challenges and limitations that require careful analysis for effective implementation. The main challenges are related to technical requirements, data privacy, and ethical aspects accompanying the use of the healthcare system in the healthcare sector.

The system requires a stable Internet connection and high-quality hardware, including high-resolution cameras, to ensure image clarity and analysis accuracy. In addition, powerful computing resources are needed to process video streams in real time.

Despite the high efficiency of computer vision and deep learning models, there is a possibility of false positives or negatives, which can negatively affect user confidence in the system. For example, challenging lighting conditions, the appearance of foreign objects in the frame, or unusual patient postures can lead to errors in incident recognition.

An important aspect is ensuring data privacy and security, as the platform processes sensitive medical information subject to strict protection standards, including GDPR and HIPAA. Even with the security measures described above, the risk of data leakage remains, especially in cyberattacks or technical failures.

In addition to technical and legal challenges, the platform faces ethical issues related to the collection and use of patient data. Implementation of the system requires the consent of patients or their guardians to store and process personal data and provide clear information about the purposes and limits of the use of information.

Improving solutions to these challenges is the basis for developing the platform and increasing the efficiency of remote patient monitoring, which will ultimately improve the quality of healthcare services and increase patient safety.

5. Conclusions

The proposed mobile video platform for diagnosing movement anomalies represents an innovative approach to monitoring patients' physical condition, providing automated detection of dangerous movement anomalies. The combination of computer vision algorithms, deep learning, and contextual analysis allows the platform to effectively recognize critical situations and inform medical professionals or decision-makers in real-time. The system's main advantages include high adaptability, the ability to integrate into patients' home environments, and the use of artificial intelligence technologies to make informed decisions. This makes the platform a promising tool for supporting patients who need constant monitoring, including the elderly or people with disabilities.

Future development of the platform may include integrating new sensors to improve the quality of data collection, optimizing algorithms for operation in different lighting conditions, and adapting to the specific needs of individual user groups. Another critical area is improving the mechanisms for protecting confidential information, allowing the platform to meet the highest security standards. The platform also has the potential to expand opportunities in telemedicine and remote care, providing more comprehensive patient support and the ability to monitor their condition effectively.

Thus, the mobile video platform for diagnosing movement abnormalities is an innovative solution for remote patient monitoring that combines high accuracy, adaptability, and ease of use. Its implementation can be an essential step in developing modern patient-centered healthcare, particularly in the context of telemedicine and supporting the quality of life of older people and people with special needs.

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